

Conformal Correlation Counterexamples

Peter Cotton

`peter.cotton@microprediction.com`

September 2026

Abstract

Perlo et al. (2026) correlate the membership indicators of conformal prediction sets and read the result as a map of which classes a classifier confuses. It is not. Two classifiers built inside randomized adaptive prediction sets, exactly as specified, produce the same law of sets and the same matrix while confusing every class in opposite directions, and a short theorem shows that no statistic of the sets determines the confusion matrix once any set of size two occurs. Seven further hand calculations give a zero entry beside 99% directional confusion, an entry of one beside one in a million co-occurrence, and an entry of ± 0.64 from pooling two clients with zero correlation on each. For classes that never share a set the entry is a function of the inclusion rates alone, and when sets have at most one element the matrix reveals a client's class proportions exactly, which undoes the privacy claim made for it.

1 The statistic

There are H classes. A classifier maps an input X to a probability vector $f(X)$, with $f_h(X)$ the probability it assigns to class h . Split conformal prediction (Vovk et al., 2005; Lei et al., 2018) turns the probability vector into a prediction set $C(X) \subseteq \{1, \dots, H\}$. The true label is Y and the class the classifier ranks first is $\hat{Y}$.

The authors build their sets with the adaptive prediction sets (APS) of Romano et al. (2020). APS assigns a score to each label at each input. Sort the probabilities at x in decreasing order, $f_{(1)}(x) \geq \dots \geq f_{(H)}(x)$, write $r(x, y)$ for the rank of class y in that order, and write $S_k(x) = \sum_{l \leq k} f_{(l)}(x)$ for the probability mass through rank k . The score of label y at input x is

$$s(x, y, u) = S_{r(x, y)}(x) - u f_y(x), \quad u \sim \text{Unif}(0, 1), \quad (1)$$

the mass at and above y in the ranking, less a uniform share of the mass of y itself.

The score is turned into a set by calibration. Given m labeled calibration pairs (x_i, y_i) with independent uniforms u_i , let $k = \lceil (m+1)(1-\alpha) \rceil$ and let τ be the k -th smallest of the calibration scores $s(x_i, y_i, u_i)$. If $\alpha < 1/(m+1)$ then $k = m+1$ and no such score exists, in which case $\tau = +\infty$ and the set is every class. We assume $\alpha \geq 1/(m+1)$ throughout, so that $k \leq m$. The prediction set at a new input x , with a fresh uniform U , is

$$C(x) = \{y : s(x, y, U) \leq \tau\}. \quad (2)$$

In words, the set is the top ranked classes down to the first rank at which the cumulative mass reaches τ , and the class at that boundary rank is kept or dropped according to U .

Two coverage bounds hold. Under exchangeability of the calibration pairs and the test pair, $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$. When the scores are almost surely distinct, also $\mathbb{P}(Y \in C(X)) \leq 1 - \alpha + 1/(m+1)$

(Romano et al., 2020, Theorem 1). Setting $u = 0$ throughout gives the non randomized variant, which is conservative. Examples 1 and 9 use the randomized variant exactly as defined here. Nothing else below depends on APS, beyond the fact that $C(X)$ is a set of classes.

The matrix is built from membership indicators. For each class h define

$$z_h = \mathbf{1}\{h \in C(X)\},$$

and write $p_h = \mathbb{E}[z_h]$ for the inclusion rate of class h and $q_{ij} = \mathbb{E}[z_i z_j]$ for the co-occurrence rate of classes i and j . The conformal correlation matrix (CCM) of Perlo et al. (2026) has entries

$$\rho_{ij} = \frac{\mathbb{E}[z_i z_j] - \mathbb{E}[z_i] \mathbb{E}[z_j]}{\sqrt{\text{Var}(z_i) \text{Var}(z_j)}} = \frac{q_{ij} - p_i p_j}{\sqrt{p_i(1-p_i) p_j(1-p_j)}}. \quad (3)$$

This is the Pearson correlation of two Bernoulli variables, also called the phi coefficient (Yule, 1912). In practice the expectations are sample averages over a test set.

The entry is defined only when $0 < p_i < 1$ and $0 < p_j < 1$, since otherwise a variance is zero. The authors give no convention for that case. We exclude such classes from the matrix.

Nothing below depends on the name or on the paper. The results concern any correlation between membership indicators of prediction sets, however the sets were built and whatever the statistic is called. The paper is the instance in print, and the move it makes, correlating the indicators and reading the result as class similarity, is cheap enough that it will be made again.

The authors attach three readings to the entries. Their Remark 1 says that intrinsically similar classes “might be expected to exhibit high positive correlation”. Their Remark 2 says that pairs of dissimilar classes “are expected to display strong negative correlation”. Their Remark 3 says that classes “unrelated or independent with respect to the classifier representation are expected to yield correlation coefficients close to zero”.

On the strength of these readings they draw three conclusions. A positive entry between two classes means the classifier “is much more likely to misclassify” one as the other. The matrix “contains a superset of the information conveyed by the confusion matrix”. And the matrix “captures the likelihood of class pairs co-occurring within conformal prediction sets”. Each of the three is false, and the examples below show it.

2 The matrix depends only on the law of the sets

Remark 1. The conformal correlation matrix is a functional of the law of the prediction set $C(X)$ alone. Two situations with the same law of $C(X)$ have the same matrix, whatever the joint law of $(C(X), Y, \hat{Y})$.

Proof. Every quantity in (3) is an expectation of a function of the vector $(z_1, \dots, z_H)$, and that vector is a function of $C(X)$. $\square$

Confusion is a different object. The confusion matrix has entries $\mathbb{P}(Y = a, \hat{Y} = b)$, which depend on the joint law of the true label and the ranked prediction. Hold the law of $C(X)$ fixed and change Y or $\hat{Y}$, and by Remark 1 the matrix cannot notice.

The remark also disposes of the claim that the entries “incorporate the entire distribution of predicted probabilities”. At a fixed α , any change to the probability vectors that leaves the sets unchanged is invisible to the matrix. The sets are a function of the probabilities and one threshold, and the matrix sees the sets.

The marginal coverage guarantee ties $C(X)$ to Y through one scalar constraint, and a construction may impose more. Under exchangeability of calibration and test data, split conformal prediction guarantees

$$\mathbb{P}(z_Y = 1) = \sum_y \mathbb{P}(Y = y, z_y = 1) \geq 1 - \alpha. \quad (4)$$

This is one number. It constrains the sum on the right and nothing else.

In particular, coverage does not in general determine a co-occurrence rate q_{ij} . That rate mixes the true class with a wrong class, two wrong classes, and different true classes on different inputs, and coverage never justifies reading it as confusion. There is one special case worth stating. With only two classes and coverage one the set is never empty, so $0 = \mathbb{P}(C = \emptyset) = 1 - p_A - p_B + q_{AB}$ and $q_{AB} = p_A + p_B - 1$ is forced. With three or more classes nothing of the kind holds, as the next lemma shows for a single entry.

Lemma 1. *Among laws of prediction sets that satisfy the marginal coverage lower bound (4), for every $\rho \in [-1, 1]$ there is one with coverage one and accuracy one whose matrix has an off-diagonal entry equal to ρ .*

Proof. Let the true label always be T , always in the set and always ranked first, so coverage and accuracy are one. Since $z_T \equiv 1$ the entries involving T are undefined, and the claim concerns ρ_{AB} . Let two other labels A and B be added to the set according to

$$\begin{aligned} \mathbb{P}(z_A = 0, z_B = 0) &= \mathbb{P}(z_A = 1, z_B = 1) = \frac{1 + \rho}{4}, \\ \mathbb{P}(z_A = 1, z_B = 0) &= \mathbb{P}(z_A = 0, z_B = 1) = \frac{1 - \rho}{4}. \end{aligned}$$

Then $p_A = p_B = \frac{1+\rho}{4} + \frac{1-\rho}{4} = \frac{1}{2}$, so each variance is $\frac{1}{4}$, and $q_{AB} = \frac{1+\rho}{4}$. Substituting into (3),

$$\rho_{AB} = \frac{(1 + \rho)/4 - 1/4}{1/4} = \rho. \quad \square$$

So every value on the correlation scale is available with no classification problem between A and B at all.

The same reasoning gives a no-go result that covers the conformal correlation matrix and any other statistic computed from the prediction sets.

Theorem 1 (No statistic of the sets determines the confusion matrix). *Consider joint laws of $(Y, \hat{Y}, C(X))$ in which $\hat{Y} \in C(X)$ whenever the set is nonempty, as under APS. Suppose some set of size at least two occurs with positive probability. Then there are two such joint laws with the same joint law of $(Y, C(X))$, hence the same law of sets, the same class proportions, the same coverage and the same value of every functional g of the law of sets, whose confusion matrices differ.*

Proof. Let S be a set with $|S| \geq 2$ and $\mathbb{P}(C(X) = S) = \delta > 0$, and let $b \neq b'$ be two of its elements. Take any admissible joint law of $(Y, \hat{Y}, C(X))$ with the given law of sets. Define a second joint law that agrees with the first except on the event $\{C(X) = S\}$, where it sets $\hat{Y} = b$, and a third that agrees with the first except on that event, where it sets $\hat{Y} = b'$. The three laws have the same joint law of $(Y, C(X))$, hence the same law of sets, the same class proportions, the same coverage $\mathbb{P}(Y \in C(X))$, and the same value of every functional g of the law of sets. Pick a class a with $\mathbb{P}(Y = a, C(X) = S) = \delta_a > 0$. The second law has confusion entry $\mathbb{P}(Y = a, \hat{Y} = b)$ larger by δ_a than the third, and entry $\mathbb{P}(Y = a, \hat{Y} = b')$ smaller by δ_a . So the second and third confusion matrices differ. $\square$

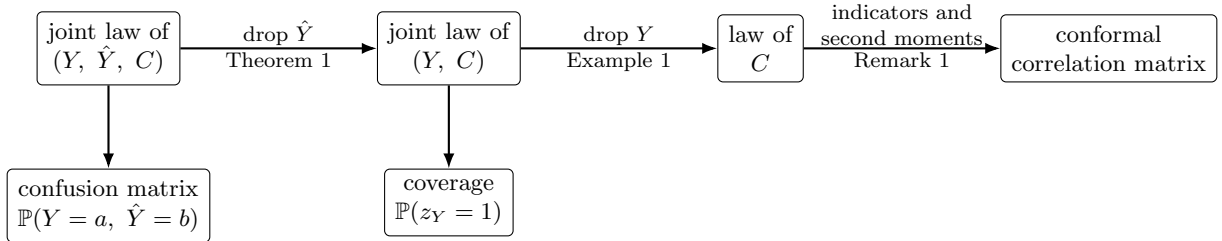


Figure 1: The chain of coarsenings from the data to the matrix. The confusion matrix is a function of the first object. Theorem 1 shows it is already lost at the first step, where the ranked prediction is dropped. Example 1 shows a further loss of one bit at the second step, where the true label is dropped. The coverage guarantee constrains one scalar of the second object. The matrix is a function of the third.

The conclusion is stronger than the title. Even the entire joint law of $(Y, C(X))$ does not determine the confusion matrix, because both constructed laws preserve it. What is lost is the information about the rankings, not only the information about the true label. Figure 1 places the two losses on the chain of coarsenings that leads from the data to the matrix.

Two limits on the theorem. First, its hypothesis is sufficient and not necessary. If every set is a singleton and coverage is one, then $\hat{Y} = Y$ and the confusion matrix is the diagonal matrix of the class proportions. With two classes, singleton sets and coverage κ , the row sums, the column sums and the trace determine all four entries. The theorem says nothing about cases in which no set of size two occurs. Second, the construction changes $\hat{Y}$ while keeping $C(X)$, and that need not be realizable by an APS classifier with its calibration, so the theorem does not establish APS realizability for every prescribed set law. Example 1 supplies a pair that is realized by two randomized APS classifiers.

The examples that follow are more concrete, and each contradicts a specific sentence in the paper.

3 Eight examples

Each example needs only the inclusion rates p_i and the co-occurrence rates q_{ij} , substituted into (3).

Examples 2 through 8 prescribe a law of prediction sets directly. By Remark 1 the law of the sets is all the matrix depends on, so a law is all an example needs to specify.

In those examples coverage is one, meaning that every set contains the true class. Any marginal coverage lower bound is then met. We do not claim that randomized APS produces these laws. It cannot produce coverage one: when $k \leq m$ and the scores are distinct, its coverage is exactly $k/(m+1) < 1$ (Romano et al., 2020).

Therefore Examples 2 through 8 refute the authors' readings of the matrix as claims about laws of prediction sets that meet the coverage bound. A claim restricted to sets produced by APS needs a counterexample built inside APS. Example 1 is that counterexample, which is why it comes first. The remaining examples are ordered from the simplest to refute and the most absurd result downward.

In Examples 4 and 6 a label T is in every set. Its own entries are undefined under the convention of Section 1, and the claims concern ρ_{AB} only.

Example 1 (Opposite confusions, the same matrix, within randomized APS). Three classes with equal frequency. Two classifiers assign the probabilities $(0.6, 0.3, 0.1)$ to the classes in the orders shown, so the true class is always ranked second.

True class	Model 1 order	Model 2 order
A	$C > A > B$	$B > A > C$
B	$A > B > C$	$C > B > A$
C	$B > C > A$	$A > C > B$

The two models get the same threshold. By (1) the score of the true label is $0.9 - 0.3u$ under both models, uniform on $(0.6, 0.9)$. Calibrate both on the same inputs with the same uniforms, with $k \leq m$ as assumed in Section 1, and both have the same threshold $\tau \in (0.6, 0.9)$. Write $r = (\tau - 0.6)/0.3$.

The two models produce sets with the same law. At a test input the top ranked class has score $0.6 - 0.6U \leq \tau$ and is always included. The second has score $0.9 - 0.3U$, which is at most τ exactly when $U \geq (0.9 - \tau)/0.3$, an event of probability r . The third has score $1 - 0.1U \geq 0.9 > \tau$ and is never included. Under either model, therefore, each singleton $\{A\}$, $\{B\}$, $\{C\}$ occurs with probability $(1 - r)/3$ and each pair $\{A, B\}$, $\{B, C\}$, $\{C, A\}$ with probability $r/3$. The same law of sets gives the same matrix. For the record, $p_i = (1 + r)/3$ and $q_{ij} = r/3$ for every class and pair, so every off-diagonal entry is

$$\rho_{ij} = \frac{r/3 - (1 + r)^2/9}{(1 + r)(2 - r)/9} = -\frac{r^2 - r + 1}{(1 + r)(2 - r)},$$

the same for both models and negative for every r .

Coverage is the same for both models too. Conditional on the calibration sample, coverage equals r exactly. Since τ is the k -th smallest of m uniforms on $(0.6, 0.9)$, the variable r is the k -th smallest of m uniforms on $(0, 1)$, which has the Beta($k, m + 1 - k$) law with mean $k/(m + 1) \geq 1 - \alpha$. So on a given calibration sample r can fall below $1 - \alpha$, and its average over calibration samples meets the marginal guarantee, as in Theorem 1 of Romano et al. (2020). The equality of the two set laws holds for every realized τ .

The two models confuse classes in opposite directions. Model 1 labels every A as C , every B as A and every C as B . Model 2 runs the cycle the other way. Their confusion matrices are transposes of each other, both with accuracy zero.

The authors write that the matrix “contains a superset of the information conveyed by the confusion matrix”. Here the sets are produced by randomized APS exactly as specified, the matrix is the same for both models, and the confusion matrices differ. The matrix does not contain the confusion matrix.

The loss can be measured. Fix the threshold and choose between the two models by a fair coin D . The set law does not depend on D , so $I(D; C) = 0$. But when $Y = A$, model 1 produces $\{C\}$ or $\{A, C\}$ and model 2 produces $\{B\}$ or $\{A, B\}$, and likewise for the other two classes, so the set together with its true label identifies D and $I(D; Y | C) = \log 2$. The sets alone carry no information about the direction of confusion. The sets with their labels carry one full bit. The matrix is computed from the sets alone. The loss is of the same kind as the one priced in Cotton (2026), where pooling a conditional residual law into one marginal law costs exactly the mutual information between residual and input. There the pooled object is the residual law and the cost is a forecasting regret. Here the pooled object is the joint law of set, label and ranked prediction, reduced to the law of the set, and the cost is a diagnostic one.

Example 2 (A perfect classifier and a perfectly wrong one share a matrix). This example makes the same point without the procedure, and with the accuracy gap as wide as it can be. Three classes A, B, C with equal frequency, and sets that always contain the true class and one neighbor.

True class	Prediction set	Good model ranks first	Bad model ranks first
A	$\{A, B\}$	A	B
B	$\{B, C\}$	B	C
C	$\{C, A\}$	C	A

Each label is in two of the three sets, so every $p_i = 2/3$ and every variance is $2/9$. Each pair shares exactly one of the three sets, so every $q_{ij} = 1/3$. Every off-diagonal entry is

$$\rho_{ij} = \frac{1/3 - (2/3)(2/3)}{2/9} = -\frac{1}{2}.$$

The good model has accuracy one and the bad model has accuracy zero. The labels and the sets are identical for both, so the matrix is identical for both. The confusion matrix $\mathbb{P}(Y = a, \hat{Y} = b)$ of the good model has all its mass on the diagonal and that of the bad model has none there. The matrix does not contain “a superset of the information conveyed by the confusion matrix”.

Example 3 (Zero correlation with 99% directional confusion). Three classes and a small number ε , say 0.01. Inputs arrive in four kinds.

Probability	True class	Prediction set	Ranked first
$(1 - \varepsilon)/2$	A	$\{A, B\}$	B
$\varepsilon/2$	A	$\{A\}$	A
$(1 - \varepsilon)/2$	B	$\{B\}$	B
$\varepsilon/2$	C	$\{C\}$	C

A is in the sets of the first two kinds, so $p_A = (1 - \varepsilon)/2 + \varepsilon/2 = 1/2$. B is in the sets of the first and third kinds, so $p_B = 1 - \varepsilon$. They appear together only in the first kind, so $q_{AB} = (1 - \varepsilon)/2 = p_A p_B$. The numerator of (3) is zero, so

$$\rho_{AB} = 0.$$

For Bernoulli indicators that are not constant, zero correlation is independence. So z_A and z_B are independent here. Yet among inputs whose true class is A , the classifier ranks B first with probability $1 - \varepsilon$, which is 99% at $\varepsilon = 0.01$.

The authors’ Remark 3 says that unrelated classes give an entry near zero. A reader of a heat map uses the converse, that an entry near zero means the classes are unrelated. The converse fails. Independence of set membership does not imply absence of classification confusion.

Example 4 (Perfect correlation with one in a million co-occurrence). Every set contains the true class T , ranked first. Two other labels A and B are added to the set together with probability ε , and otherwise neither is added. Then $z_A = z_B$ on every input, so

$$\rho_{AB} = 1 \quad \text{for every } 0 < \varepsilon < 1.$$

The probability that A and B appear together in a set is $q_{AB} = \varepsilon$, which may be 0.4, or 0.01, or one in a million. The authors write in their abstract that the matrix “captures the likelihood of class pairs co-occurring”. The likelihood is ε . The entry is one regardless.

Example 5 (Every pair positive from difficulty alone). Three classes with equal frequency, and difficulty independent of the true class. Half the inputs are hard and receive the set of all three

classes. The other half are easy and receive the singleton set of their true class. The true class is ranked first in both cases, so accuracy is one.

Each label is in every hard set and in a third of the easy sets, so

$$p_i = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{3} = \frac{2}{3}, \quad p_i(1 - p_i) = \frac{2}{9}.$$

Two labels appear together only in hard sets, so $q_{ij} = 1/2$ for every pair, and

$$\rho_{ij} = \frac{1/2 - (2/3)(2/3)}{2/9} = \frac{1}{4}.$$

Every pair is positively correlated and the classifier is perfect. The correlation has one cause, whether the input was hard, and not from any relation between the pair. The authors' Remark 1 reads a positive entry as "intrinsically similar classes".

Example 6 (Pooling two clients creates a strong correlation from nothing). Every set contains the true class T , ranked first. On client 1 the labels A and B are each added to the set independently with probability 0.9. On client 2 they are each added independently with probability 0.1. The clients contribute equally to the pooled data.

Within each client z_A and z_B are independent, so $\rho_{AB} = 0$ on each. Pooled, $p_A = p_B = (0.9 + 0.1)/2 = 1/2$, each variance is $1/4$, and

$$q_{AB} = \frac{1}{2}(0.9 \times 0.9) + \frac{1}{2}(0.1 \times 0.1) = 0.41, \quad \rho_{AB} = \frac{0.41 - 1/4}{1/4} = 0.64.$$

The sign can be reversed. Let client 1 add A with probability 0.9 and B with probability 0.1, and let client 2 do the opposite. Then $q_{AB} = \frac{1}{2}(0.09) + \frac{1}{2}(0.09) = 0.09$ and $\rho_{AB} = (0.09 - 0.25)/0.25 = -0.64$.

The authors compute matrices across clients that are non identically distributed by construction and read the entries as properties of the classifier. A strong entry of either sign is produced here by the mixture of clients alone, with no confusion anywhere.

Example 7 (Two classes the model never confuses get different entries). Three classes, cat, dog and plane, with equal frequency. The classifier ranks the true class first on every input and returns the same set for every input of a given class.

True class	Prediction set	z_{cat}	z_{dog}	z_{plane}
cat	{cat, dog}	1	1	0
dog	{dog}	0	1	0
plane	{plane}	0	0	1

A third of the inputs are cats, so $p_{\text{cat}} = 1/3$, and likewise $p_{\text{dog}} = 2/3$ and $p_{\text{plane}} = 1/3$. Each variance is $2/9$. Cat and dog appear together on the cat inputs, so $q_{\text{cat,dog}} = 1/3$. Plane never appears with either, so the other two co-occurrence rates are zero. Substituting,

$$\begin{aligned} \rho_{\text{cat,dog}} &= \frac{1/3 - (1/3)(2/3)}{2/9} = \frac{1}{2}, \\ \rho_{\text{cat,plane}} &= \frac{0 - (1/3)(1/3)}{2/9} = -\frac{1}{2}, \\ \rho_{\text{dog,plane}} &= \frac{0 - (2/3)(1/3)}{2/9} = -1. \end{aligned}$$

The authors' Remark 2 reads this as dog and plane being strongly dissimilar and cat and plane half as dissimilar. But the classifier treats plane identically with respect to both. It never puts plane in a set with either. The two entries differ only because dog is in twice as many sets as cat. Lemma 2 gives the general statement.

Example 8 (Moving one label occurrence between rows moves an entry from one to almost zero). There are n test inputs. Every set contains the true class T , ranked first, and two rare labels A and B each appear in exactly one set. If they appear in the same set then $z_A = z_B$ and $\rho_{AB} = 1$. If instead the one occurrence of B is in a different set then $p_A = p_B = 1/n$, $q_{AB} = 0$, and

$$\rho_{AB} = \frac{0 - 1/n^2}{(1/n)(1 - 1/n)} = -\frac{1}{n - 1},$$

which at $n = 1000$ is about -0.001 .

The authors present heat maps of entries with no uncertainty attached. For rare labels, moving a single label occurrence from one row to another moves an entry from one to about zero, half of its range.

4 Two lemmas

Write $o_i = p_i/(1 - p_i)$ for the odds of inclusion of class i .

Lemma 2 (The floor, and the attainable range). *If classes i and j never share a set, then*

$$\rho_{ij} = -\sqrt{o_i o_j} = -\sqrt{\frac{p_i p_j}{(1 - p_i)(1 - p_j)}}. \quad (5)$$

With H classes of equal inclusion rate and singleton sets this equals $-1/(H - 1)$. More generally, for fixed inclusion rates $p_i \leq p_j$ the entry ρ_{ij} ranges over

$$\left[-\sqrt{o_i o_j}, \sqrt{o_i o_j} \right] \quad \text{if } p_i + p_j \leq 1, \quad \left[-1/\sqrt{o_i o_j}, \sqrt{o_i o_j} \right] \quad \text{if } p_i + p_j > 1.$$

Proof. If $q_{ij} = 0$ then (3) reads

$$\rho_{ij} = \frac{-p_i p_j}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{p_i^2 p_j^2}{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{p_i p_j}{(1 - p_i)(1 - p_j)}}.$$

With singleton sets every pair has $q_{ij} = 0$, and with equal inclusion rates $p_i = 1/H$, so $o_i = 1/(H - 1)$ and $\rho_{ij} = -1/(H - 1)$.

For the range, q_{ij} is the probability of the intersection of two events with probabilities $p_i \leq p_j$, so by the Fréchet bounds $\max(0, p_i + p_j - 1) \leq q_{ij} \leq p_i$. At the upper bound the numerator of (3) is $p_i - p_i p_j = p_i(1 - p_j)$ and

$$\rho_{ij} = \frac{p_i(1 - p_j)}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = \sqrt{\frac{p_i(1 - p_j)}{(1 - p_i)p_j}} = \sqrt{o_i o_j}.$$

At the lower bound, if $p_i + p_j \leq 1$ then $q_{ij} = 0$ and (5) applies. If $p_i + p_j > 1$ then $q_{ij} = p_i + p_j - 1$, the numerator is $p_i + p_j - 1 - p_i p_j = -(1 - p_i)(1 - p_j)$, and

$$\rho_{ij} = \frac{-(1 - p_i)(1 - p_j)}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{(1 - p_i)(1 - p_j)}{p_i p_j}} = -1/\sqrt{o_i o_j}. \quad \square$$

The lemma has two consequences for reading a heat map. The first concerns the scale. An entry reads +1 only when $p_i = p_j$, so a rare class and a common class have a lower ceiling than two common classes, and entries for pairs with different inclusion rates are not on a common scale.

The second concerns α . With the model, the calibration data and the auxiliary uniforms fixed, the APS sets are nested in α . As α grows no inclusion rate increases, so every floor entry (5) moves toward zero or stays put, whether or not the classifier changed, as long as the indicators involved remain nondegenerate. The authors sweep α over four values and present the dependence as a feature.

Lemma 3 (Identification of the inclusion rates). *Suppose every prediction set has at most one element and at least three classes have inclusion rate strictly between zero and one. Then for any three distinct such classes i, j, k ,*

$$o_i = -\frac{\rho_{ij} \rho_{ik}}{\rho_{jk}}, \quad p_i = \frac{o_i}{1 + o_i}. \quad (6)$$

The off-diagonal entries of the matrix therefore determine every inclusion rate exactly.

Proof. With sets of at most one element no two classes ever share a set, so (5) holds for every pair, and $\rho_{ij} \rho_{ik} = \sqrt{o_i o_j} \sqrt{o_i o_k} = o_i \sqrt{o_j o_k} = -o_i \rho_{jk}$. Divide by $-\rho_{jk}$, which is nonzero since $o_j, o_k > 0$. The map $o \mapsto o/(1 + o)$ inverts $p \mapsto p/(1 - p)$. $\square$

The authors argue that the matrix is privacy preserving because it “only contains probabilities” and so “does not leak information on the size and composition” of a client’s data. Correlations can be negative, so the matrix does not contain probabilities. A row normalized confusion matrix does contain only probabilities, and does not directly disclose counts either.

Lemma 3 says more. When every set has at most one element, the off-diagonal entries recover every inclusion rate. Inside randomized APS that recovers the class proportions exactly when every class is retained with the same probability, which APS does not guarantee in general but which the next construction arranges.

Example 9 (The matrix reveals the class proportions, within randomized APS). Take a perfect classifier that assigns the ranked probabilities $(0.6, 0.3, 0.1)$ with the true class always first, and let the true class proportions be π_1, π_2, π_3 , all positive. By (1) every calibration score is $0.6 - 0.6u$, so $\tau \in (0, 0.6)$. At a test input the top class has score $0.6 - 0.6U \leq \tau$ exactly when $U \geq 1 - \tau/0.6$, an event of probability $r = \tau/0.6$, and the second has score $0.9 - 0.3U \geq 0.6 > \tau$ and is never included. So the set is $\{Y\}$ with probability r and empty otherwise, every set has at most one element, and $p_i = r \pi_i \in (0, 1)$. Lemma 3 recovers every p_i from the off-diagonal entries, and then

$$\pi_i = \frac{p_i}{\sum_j p_j}.$$

The population matrix determines the true class proportions exactly. For an empirical matrix the same algebra recovers the empirical inclusion frequencies exactly, provided at least three classes are nondegenerate in the sample and the entries are not rounded, and the proportions are then estimated rather than read off.

The matrix is offered as useful because it exposes how a client’s data differs from the others. No threat model, mechanism or leakage bound is given.

5 What the coverage guarantee constrains

The authors state that “for a given input sample” the set “is guaranteed to contain the correct class” with probability close to $1 - \alpha$. The guarantee (4) is marginal. It does not imply coverage conditional on the input, on a class or on a client. In distribution free regression with nonatomic covariates, exact pointwise conditional coverage forces essentially trivial prediction sets of infinite expected length (Barber et al., 2021). The client and class example below makes the classification point directly.

The distinction matters here because the authors interpret individual class pairs and individual federated clients as if validity transferred to them. It does not. Take ten equally weighted clients, nine with coverage one and one with coverage zero. Pooled coverage is exactly 0.9, which meets a nominal $1 - \alpha = 0.9$, and the tenth client has no protection at all. The same arithmetic applies to a class with a tenth of the data that is missed every time.

The authors also write the upper coverage term as $1/(n - 1)$. For the randomized procedure with almost surely distinct scores the term is $1/(n + 1)$ (Romano et al., 2020; Lei et al., 2018), so theirs is looser than necessary and probably a typographical error.

The federated experiments compare local matrices computed against a global calibration set across clients that are non identically distributed by construction. Under that mismatch even the marginal guarantee need not hold. No coverage or set sizes are reported for any client.

6 What to report instead

If the question is which classes the classifier confuses, the probabilities answer it directly. The soft confusion matrix

$$C_{ab} = \mathbb{E}[f_b(X) \mid Y = a]$$

is the average probability assigned to class b on inputs of true class a . It is directional and threshold free, and it needs no calibration set. It needs labeled evaluation data, as any confusion matrix does.

The matrix C_{ab} is a conditional mean, so it summarizes rather than determines. A model with $f_B(X) = 0.4$ on every input of class A and a model with $f_B(X)$ equal to 0.6 or 0.2 with equal probability have the same row (0.6, 0.4), and error rates from A to B of zero and one half. The confusion matrix of the ranked prediction carries that difference and C_{ab} does not, so the two are complements.

If labels are unavailable, the mean pairwise product $\mathbb{E}[f_i(X)f_j(X)]$ is symmetric and needs no labeled calibration or evaluation sample. If sets are the object, because a downstream consumer receives sets and nothing else, then the co-occurrence rate q_{ij} is the quantity the abstract describes, and the coverage confusion matrix of Stutz et al. (2022), which stratifies by the true label, is directional. Its row normalized form $\mathbb{P}(k \in C(X) \mid Y = y)$ is the conditional version.

In every case the inclusion rates and the set sizes should be reported alongside, since by Lemma 2 an entry of (3) cannot be read without them.

The authors compare against none of these. Their only comparison is with the hard confusion matrix, which discards the probabilities entirely, so the contrast they draw is soft versus hard rather than conformal versus not.

The authors’ client selection experiment does not bear on any of this, but we note four things. It selects the clients with the highest entries for two critical classes, while the centralized section reads a falling entry as the classifier learning to separate them. It generates heterogeneity with a symmetric Dirichlet($\beta, \dots, \beta$) and states that larger β gives more heterogeneous clients, whereas $\text{Var}(\theta_h) = (H - 1)/(H^2(H\beta + 1))$ decreases in β . Its Figures 6 and 7 are captioned as matrix entries

while their axes are class accuracy, and its Table 1 lists six selected CIFAR-10 clients where the text says seven. The same table reports accuracy on the non critical classes moving from 0.513 to 0.494 on CIFAR-100 and from 0.859 to 0.856 on CIFAR-10, and overall CIFAR-100 accuracy from 0.511 to 0.493, under the conclusion that the method “does not affect” non critical performance, without repetitions, seeds or error bars.

Simulations of Example 1, of Examples 5 and 7, and of the direction blindness of Example 2, on a synthetic ten class classifier with adaptive prediction sets and with C_{ab} and $\mathbb{E}[f_i f_j]$ alongside as α varies, are available at <https://conformalprediction.net/demos/32-conformal-correlation.html>.

7 Conclusion

The conformal correlation matrix is the phi coefficient of APS set membership indicators. By Remark 1 it sees only the law of the sets, by Lemma 1 the marginal coverage lower bound leaves any single entry free among the laws that satisfy it, and by Theorem 1 no statistic of the sets, this one or any other, determines the confusion matrix once a set of size at least two occurs.

Example 1 shows, inside randomized APS itself, that the matrix does not determine the confusion matrix. Examples 2 through 8 concern laws of sets that meet the coverage bound. Among them, Example 3 disproves the converse of the authors’ Remark 3, Example 4 shows that an entry does not determine the co-occurrence probability, and Examples 5 and 6 show positive entries with no pairwise confusion. Lemma 2 shows that for pairs that never share a set the entry is a function of the inclusion rates alone, and Lemma 3 with Example 9 shows that inside randomized APS a perfect classifier’s matrix reveals the class proportions. None of this is repaired by the certificate, which concerns the true label and does not generally determine which other labels sit beside it.

References

- R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. The limits of distribution-free conditional predictive inference. *Information and Inference*, 10(2):455–482, 2021.
- P. Cotton. Marginally useful: an information-gap identity in conformal prediction. arXiv:2608.07479, 2026.
- J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- A. Perlo, C. F. Chiasserini, G. De Veciana, and F. Malandrino. Characterizing the performance of classification models through conformal correlation matrices. *Computer Communications*, 247:108398, 2026.
- Y. Romano, M. Sesia, and E. J. Candès. Classification with valid and adaptive coverage. In *Advances in Neural Information Processing Systems*, volume 33, pages 3581–3591, 2020.
- D. Stutz, K. Dvijotham, A. T. Cemgil, and A. Doucet. Learning optimal conformal classifiers. In *International Conference on Learning Representations*, 2022.
- V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

G. U. Yule. On the methods of measuring association between two attributes. *Journal of the Royal Statistical Society*, 75(6):579–652, 1912.